

Arnav Revankar

Junior Computer Engineer @ NJIT

github.com/avr34 | avr33@njit.edu | linkedin.com/in/avr33

EDUCATION

-
- New Jersey Institute of Technology** (Sept 2023 - Present) Cumulative GPA: 3.65
BS Computer Engineering, Minor in Data Analytics Newark, NJ
- **Coursework:** Circuits and Systems 1 & 2, Microprocessors, Computer Architecture, Applied Machine Learning, Digital Design, and others.
 - **Activities:** IEEE-HKN Inductee, 2026 Hackathon Volunteer
- County College of Morris** (Sept 2022 - May 2023) Cumulative GPA: 3.83
Challenger Program for High School Students Randolph, NJ
- Completed over two semesters as a part-time student.

EXPERIENCE

-
- Software Engineering Intern** March 2025 - August 2025
Mathnasium Chatham, NJ
- Spearheaded a project developing a full-stack Javascript/Node.js email generator for parents of students at a Mathnasium tutoring center.
 - Effectively employs a BNF Grammar, that is used to generate syntactically valid text from a set of customizable templates.
 - Currently being used to send 110-150 emails per day, saving 5 centers 30-40 hours a week.

SKILLS

-
- **Languages:** C, C++, MATLAB, Java, Python, Go, SQL (MySQL), GNU Octave, $\text{\LaTeX}$, Javascript, VHDL
 - **Software Tools:** Linux/Unix, KiCAD, FreeCAD, STM32CubeMX & IDE, PlatformIO, ESP-IDF, Git, Docker, Excel
 - **Build Tools/Debuggers:** Apache Maven, Apache Ant, CMake, Make, MSVC & GCC, GDB, Valgrind
 - **Libraries:** STM32 HAL, FreeRTOS, NumPy, PyTorch, Scikit Learn, Matplotlib, Gnuplot
 - **Communication Protocols:** UART, I2C, SPI, CAN, USB 2.0, TCP and Socket programming in C/C++ and Go

PROJECTS

-
- SolarVision Autonomous Mower** ([Github](#)) Sept 2025 - May 2025
- YOLO and LiDAR-based autonomous lawnmower, engineering capstone project. Still in progress
 - Designed a custom 6-layer PCB for STM32N6 MCU, with external SPI NOR Flash and SDRAM Memory through the FMC.
 - Trained a YOLOv8 Segmentation model on a custom dataset, and exported to C for the STM32N6 NPU, with the ST Edge AI Tool Suite.
 - LiDAR data acquisition and path planning on an ESP32S3
- 8-Channel Logic Analyzer** ([Github](#)) Feb 2026 - May 2025
- USB Logic Analyzer built around an STM32F446, capable of sniffing upto 500kHz signals.
 - DMA data acquisition, and run-length encoded data transmission, to optimize byte usage on USB 2.0 FS.
 - UART, SPI, I2C, and CAN signal decoding via a custom Go backend
 - GUI with PyQt5 in Python
- BitSum Lecture Summarizer** ([GitHub](#)) Oct 2025 - Dec 2025
- Built a lecture summarizer tool, integrating Microsoft BitNet, OpenAI Whisper, and a custom feedforward network.
 - Transcribes and summarizes lectures on CPU, 2x faster than human comprehension on average.
 - Integrated with Docker to ensure reproducibility.
- NeuralNet** ([Github](#)) Oct 2024 - Feb 2025
- Fully-Connected Neural Network Library for Regression and Classification in C.
 - Backpropagation using Stochastic Gradient Descent in Matrix form.
 - Last effort was converting to fixed-point weights, to run on resource constrained microcontrollers, TinyML style.